

CALCULUS I | AP CALCULUS AB/BC

Related Rates

A Structured Problem-Solving Framework

Author: Robbie Kohn

Level: Grades 11–12 / College Intro

| 60–75 Minutes

Prereqs: Chain Rule, Implicit
Differentiation, Geometric Formulas

Artifact: Instructional Sample

color rulegray **Central Idea:** Translating dynamic verbal scenarios into rigorous mathematical relations by treating changing geometric quantities as functions of time.

Instructional Focus & Objectives

Related rates problems represent one of the critical transitions in introductory calculus: moving from purely algebraic differentiation to genuine mathematical modeling. Rather than viewing each word problem as an ad-hoc collection of tricks, this lesson develops a repeatable, reasoned framework that guides students from verbal description to geometric relation, implicit time-differentiation, and physical interpretation.

- **Conceptual Understanding:** Explain why dynamic quantities must be treated as implicit functions of time t , and understand why numerical values cannot be substituted into geometric relationships prior to differentiation.
- **Mathematical Modeling:** Translate verbal descriptions of physical situations into clear coordinate diagrams, explicit variable definitions, and governing geometric equations.
- **Procedural Competency:** Apply the Chain Rule to differentiate multivariable relationships with respect to time (d/dt), systematically relating known rates to target unknown rates.
- **Interpretation & Justification:** Interpret the algebraic sign, magnitude, and compound dimensional units (such as m/s , m^2/s , or ft^3/min) of calculated rates within the physical context.

Anticipated Misconceptions & Pedagogical Interventions

Critical Error 1: Premature Numerical Substitution (“Freezing the Frame”)

Misconception: Substituting instantaneous numerical measurements (e.g., setting the ladder height $y = 12$ ft or circle radius $r = 10$ cm) *before* differentiating the relating equation.

Mathematical Reality: If y is set to the constant 12 before differentiation, the derivative $\frac{d}{dt}(12) = 0$. Substituting too early treats a dynamic, changing variable as a rigid constant, obliterating the rate of change $\frac{dy}{dt}$.

Pedagogical Prompt: “Is this quantity changing throughout the physical process, or is it fixed from the start? If it changes, it must remain a variable symbol until after the derivative is taken.”

Critical Error 2: Sign Neglect for Decreasing Quantities

Misconception: Leaving all rates positive because speed is colloquially described as a positive quantity (e.g., stating $\frac{dy}{dt} = 2 \text{ ft/s}$ when the top of a ladder slides downward).

Mathematical Reality: The derivative reflects directional displacement over time. If a geometric quantity decreases as time advances, its derivative must be strictly negative ($\frac{dy}{dt} < 0$). Failing to assign the negative sign causes contradictory algebraic signs in governing balance equations.

Pedagogical Prompt: “As time advances ($dt > 0$), does this distance get longer or shorter? If it shrinks, what must be the sign of the rate?”

Critical Error 3: Confusing Instantaneous State with Dynamic Rate

Misconception: Conflating the current measure of a variable (such as water depth h) with its instantaneous rate of change ($\frac{dh}{dt}$).

Pedagogical Prompt: “Check your units. Is the problem describing feet, or feet per second? A position is a state; a derivative is a velocity.”

Lesson Launch: Connecting Geometry, Time, and the Chain Rule**Teacher Guidance & Instructional Launch (7–10 Minutes)**

The Hook: Display an inflating round balloon. Ask the class: “If air enters at a steady rate of $100 \text{ cm}^3/\text{s}$, does the radius expand at a constant speed? Why or why not?”

Guiding the Insight: Solicit quick student predictions. Most students intuitively recognize that when the balloon is tiny, the radius expands rapidly, but as the balloon grows large, the same volume influx produces only a sluggish change in radius.

The Core Mathematical Link: This physical intuition embodies related rates. The quantities $V(t)$ and $r(t)$ are coupled via geometry:

$$V(t) = \frac{4}{3}\pi(r(t))^3$$

Because both quantities depend on time t , differentiating both sides with respect to t requires the **Chain Rule**:

$$\frac{d}{dt}[V] = \frac{d}{dr}[V] \cdot \frac{dr}{dt} \implies \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

The rates $\frac{dV}{dt}$ and $\frac{dr}{dt}$ are proportional, but their factor of proportionality ($4\pi r^2$, the instantaneous surface area!) changes at every instant.

A Structured Problem-Solving Framework

While related-rates problems arise across varied geometries (circles, sliding ladders, filling tanks, shadows), the underlying reasoning follows an organized progression. This framework provides students with a dependable roadmap for translating complex text into reliable calculus.

The Related Rates Problem-Solving Framework

Step 1: Analyze the Scenario & Sketch the Geometry: Draw a clear, uncluttered diagram representing the situation at a general instant t . Label all changing quantities with variable letters (x, y, r, θ, V). Label fixed constants with their numerical values.

Step 2: Catalog Variables, Instantaneous Values, and Rates: Create an explicit reference ledger:

- **Known rates:** $\frac{d[\text{var}]}{dt} = \dots$ (with appropriate signs and units)
- **Target rate:** $\frac{d[\text{var}]}{dt} = ?$ at the specific instant (e.g., when $x = x_0$).

Step 3: Formulate the Relating Equation: Write a single geometric or physical equation relating the active variables (e.g., Pythagorean theorem, trigonometric ratio, area, volume, or similar triangles). The equation should contain *only* variables whose values or rates are known or sought.

Step 4: Differentiate with Respect to Time (t): Apply implicit differentiation to both sides of the equation with respect to t . Remember that every dynamic variable is an implicit function of time:

$$\frac{d}{dt}[x^2] = 2x \frac{dx}{dt}, \quad \frac{d}{dt}[xy] = x \frac{dy}{dt} + y \frac{dx}{dt}$$

Step 5: Determine Auxiliary Instantaneous Values: If the differentiated equation involves unknown static quantities at the moment of interest (e.g., finding y when x is given), return to the *original* geometric relation to solve for them before substituting.

Step 6: Substitute Instantaneous Values & Solve for the Target Rate: Substitute all known static values and known rates into the derivative equation. Isolate the requested rate algebraically.

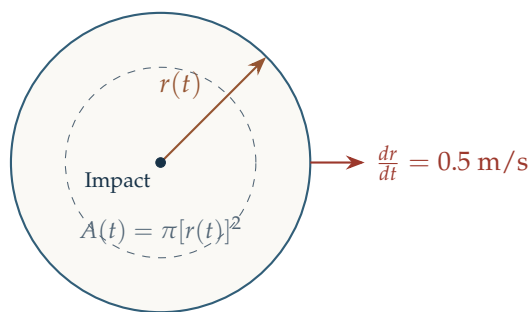
Step 7: Interpret the Result in Context: State the answer in a complete sentence, verifying physical plausibility, explaining the meaning of the algebraic sign (increasing vs. decreasing), and attaching compound units.

Guided Modeling & Worked Examples

Example 1: Expanding Circular Wave (Foundational Model)

Problem Statement

A pebble dropped into a calm pond creates an expanding circular ripple. The radius of the ripple expands outward at a constant speed of 0.5 m/s. At what rate is the enclosed circular area increasing at the instant the radius reaches 8 m?

**Step-by-Step Solution:**

Step 1: Diagram: Circle with center at impact point, dynamic radius $r(t)$, dynamic area $A(t)$.

Step 2: Given & Target: $\frac{dr}{dt} = +0.5 \text{ m/s}$. Find $\frac{dA}{dt}$ at the moment $r = 8 \text{ m}$.

Step 3: Relating Equation: $A = \pi r^2$.

Step 4: Differentiate with respect to t :

$$\frac{d}{dt}[A] = \frac{d}{dt}[\pi r^2] \implies \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Step 5: Auxiliary Values: Not required; $r = 8 \text{ m}$ is directly provided.

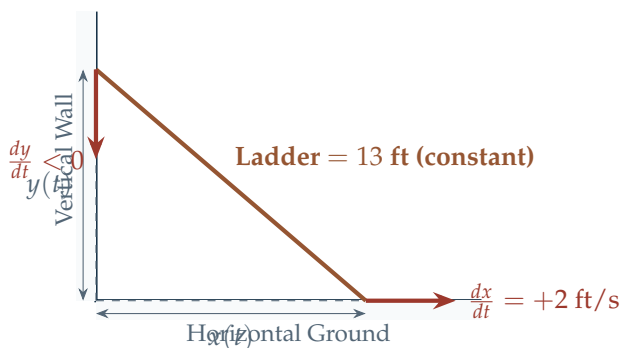
Step 6: Substitute and Solve:

$$\frac{dA}{dt} = 2\pi(8 \text{ m}) \left(0.5 \frac{\text{m}}{\text{s}}\right) = 8\pi \frac{\text{m}^2}{\text{s}} \approx 25.13 \frac{\text{m}^2}{\text{s}}$$

Step 7: Contextual Interpretation: When the ripple's radius is 8 m, the area enclosed by the ripple is expanding at exactly $8\pi \text{ m}^2/\text{s}$ (approximately $25.13 \text{ m}^2/\text{s}$). The positive sign confirms the area is growing over time.

Example 2: The Sliding Ladder (Classic Constrained Motion)**Problem Statement**

A rigid 13-foot ladder rests against a vertical exterior wall. The base of the ladder begins slipping along the level ground away from the wall at a steady speed of 2 ft/s. How fast is the top of the ladder sliding downward along the wall at the instant the base is 5 feet away from the wall?



Step-by-Step Solution:

Step 1: Diagram: Right triangle formed by vertical wall (y), horizontal ground (x), and constant hypotenuse ($L = 13$ ft).

Step 2: Catalog Quantities: Known rate: $\frac{dx}{dt} = +2$ ft/s (base moves away from origin, x increases).

Target rate: $\frac{dy}{dt} = ?$ at the instant $x = 5$ ft.

Constant: Hypotenuse $L = 13$ ft (does not depend on time; $\frac{dL}{dt} = 0$).

Step 3: Relating Equation: By the Pythagorean Theorem:

$$x^2 + y^2 = 13^2 = 169$$

Step 4: Differentiate with respect to t :

$$\frac{d}{dt}[x^2 + y^2] = \frac{d}{dt}[169] \implies 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \implies x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

Step 5: Auxiliary Instantaneous Value: We need the height y at the exact instant $x = 5$ ft. Using the geometric relation:

$$5^2 + y^2 = 169 \implies y^2 = 144 \implies y = +12 \text{ ft} \quad (y > 0 \text{ physically})$$

Step 6: Substitute and Solve:

$$(5 \text{ ft}) \left(2 \frac{\text{ft}}{\text{s}} \right) + (12 \text{ ft}) \left(\frac{dy}{dt} \right) = 0$$

$$10 + 12 \frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{10}{12} = -\frac{5}{6} \frac{\text{ft}}{\text{s}}$$

Step 7: Contextual Interpretation: The algebraic value is $\frac{dy}{dt} = -\frac{5}{6}$ ft/s. The negative sign confirms that the vertical position $y(t)$ is decreasing as time advances. In plain physical terms: the top of the ladder is sliding downward at a rate of $\frac{5}{6}$ ft/s (approximately 0.833 ft/s).

Teacher Note: Instructional Questioning During Example 2

Ask the students: “What happens as the foot of the ladder approaches 13 ft? Look at the formula $\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$. As $x \rightarrow 13$, $y \rightarrow 0$, which implies $\frac{dy}{dt} \rightarrow -\infty$. Does the ladder top really hit infinite speed?”

Discussion: A great opportunity to remind students that our mathematical model assumes the ladder maintains continuous contact with the frictionless wall. In reality, the ladder loses contact with the wall before reaching the ground. Mathematical models reflect their physical assumptions!

Class Practice & Scaffolded Exercises

- Spherical Expansion:** Helium is pumped into a spherical weather balloon at a constant rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon expanding at the instant the radius is 20 cm? ($V = \frac{4}{3}\pi r^3$)
- Two-Vehicle Separation:** Car A leaves an intersection traveling North at 30 mph. Car B leaves the same intersection traveling East at 40 mph. At what rate is the straight-line distance between them increasing 1 hour after departure?

3. **Inverted Conical Reservoir (Reduced Scaffolding):** Water flows into an inverted conical tank (vertex down) of radius 5 ft and height 10 ft at a constant rate of $2 \text{ ft}^3/\text{min}$. At what rate is the water level rising when the water is 4 ft deep?
Scaffolding prompt: Express volume $V = \frac{1}{3}\pi r^2 h$ in terms of h alone by setting up a similar triangle ratio between radius r and height h .
4. **Varying Rectangle:** A rectangle's length is increasing at 2 cm/s while its width is decreasing at 3 cm/s . At what rate is the area changing when length is 12 cm and width is 8 cm? Is the area increasing or decreasing?

Formative Assessment: Individual Exit Task

Name: _____ Date: _____

Task A (Procedural Execution): A circular oil slick spreads on the surface of water such that its radius increases at a constant rate of 0.75 m/min . Find the exact rate at which the contaminated area is expanding at the instant the diameter of the slick is 20 meters. Show all steps using the structured framework.

Task B (Conceptual Justification): A classmate attempts to solve the problem above by setting $A = \pi(10)^2 = 100\pi$ and then taking the derivative with respect to time, concluding that $\frac{dA}{dt} = 0 \text{ m}^2/\text{min}$.

- (a) Explain precisely why this classmate's reasoning is mathematically invalid.
- (b) What physical distinction is lost when numbers are substituted prior to differentiating?

Differentiation & Pedagogical Supports

Structured Support (Struggling Learners): • **Variable Ledger Graphic Organizer:** Provide a pre-printed four-column ledger: [Variable Name] | [Physical Meaning] | [Instantaneous Value] | [Rate of Change (d/dt)].

- **Color-Coded Differentiation:** Encourage students to write time-derivatives ($\frac{d}{dt}$) in a distinct color to visibly highlight where the Chain Rule introduces dynamic rates.
- **Unit Consistency Check:** Insist that every substituted number carry its physical units; if the resulting algebraic unit does not match the target rate unit, a step was omitted.

- Enrichment & Extension (Advanced Learners):**
- **Angular Rates of Change:** In the sliding ladder scenario, determine the rate at which the angle of elevation θ between the ladder and the ground is changing at the same instant ($\frac{d\theta}{dt}$).
 - **Generalized Dimensions:** Solve the conical reservoir problem for arbitrary flow rate Q , maximum radius R , and height H , showing that the surface rise rate $\frac{dh}{dt} = \frac{QH^2}{\pi R^2 h^2}$ exhibits an inverse-square dependence on instantaneous depth.

Curricular Context & References

This instructional sample aligns with standard secondary and introductory collegiate calculus curricula (College Board AP Calculus AB/BC Topic 4.4 and 4.5: Related Rates).

- Stewart, J. *Calculus: Early Transcendentals*, 8th ed.; Cengage Learning, 2016; Chapter 3.9 (Related Rates).
- Larson, R.; Edwards, B. H. *Calculus*, 11th ed.; Cengage Learning, 2018; Section 2.6.
- College Board. *AP Calculus AB and BC Course and Exam Description*, Effective Fall 2020; Unit 4: Contextual Applications of Differentiation.